



PTT PHENOL COMPANY
LIMITED

PROCESS SPECIFICATION

DECOMPOSER PRODUCT PUMPS (P-2303A/B)

Docu. No. : 14780-8120-PS-P2303
Project No. : 120117
Rev. No. : Z1
Date : MAY 10, 2016
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TITLE :

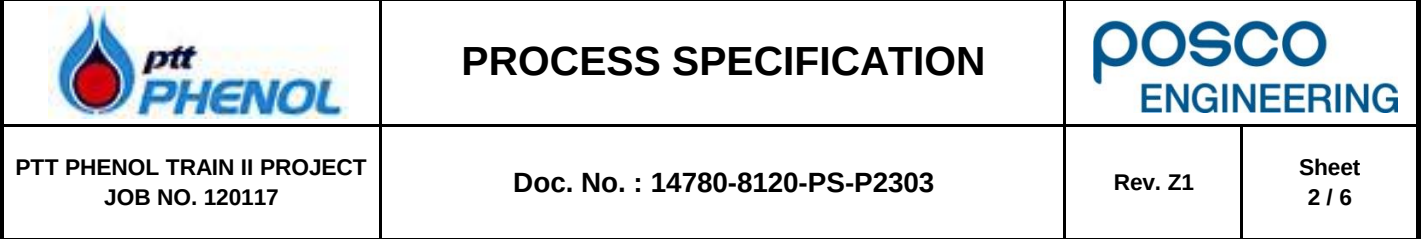
DECOMPOSER PRODUCT PUMPS

(P-2303A/B)

DISCIPLINE : **PROCESS**
DOCUMENT NO. : **14780-8120-PS-P2303**
PROJECT NAME : **PTT Phenol Train II Project**
PROJECT NO. : **120117**
OWNER : **PTT Phenol Company Limited (PPCL)**

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Record of Revision

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CENTRIFUGAL PUMPS AND DRIVERS

PROJECT SPECIFICATION

963766 - 501

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REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	09-Apr-12	MW	DRL				
1	25-Apr-12	MW	MEH				
2	14-Aug-12	MW	MEH				
3	12-Sep-12	MW	MEH				

REV	APPLICABLE TO: ☐ PROPOSALS ☐ PURCHASE ☒ AS BUILT	1
	FOR _____ UNIT _____	2
	SITE _____ SERVICE <u>Decomposer Product Pump</u>	3
	NO. REQ <u>2</u> PUMP SIZE _____ TYPE <u>Process</u> NO. STAGES <u>1</u>	4
	MANUFACTURER _____ MODEL _____ SERIAL NO. _____	5
	NOTES: INFORMATION BELOW TO BE COMPLETED: ☐ BY PURCHASER ☐ BY MANUFACTURER ☒ BY MANUFACTURER OR PURCHASER	6
	GENERAL	7
	PUMPS TO OPERATE IN (PARALLEL) _____ NO. MOTOR DRIVEN <u>2</u> NO. TURBINE DRIVEN _____	8
	(SERIES) WITH _____ PUMP ITEM NO. <u>P-2303A/B</u> PUMP ITEM NO. _____	9
	GEAR ITEM NO. _____ MOTOR ITEM NO. _____ TURBINE ITEM NO. _____	10
	GEAR PROVIDED BY _____ MOTOR PROVIDED BY _____ TURBINE PROVIDED BY _____	11
	GEAR MOUNTED BY _____ MOTOR MOUNTED BY _____ TURBINE MOUNTED BY _____	12
	GEAR DATA SHT. NO. _____ MOTOR DATA SHT. NO. _____ TURBINE DATA SHT. NO. _____	13
	OPERATING CONDITIONS (5.1.3)	14
	☐ CAPACITY, NORMAL <u>68.3</u> m³/h RATED <u>75.1</u> m³/h OTHER _____ m³/h ☐ SUCT PRESS MAX./RATED <u>11.63</u> / <u>1.314</u> kg/cm²[g]	15
	☐ DISCHARGE PRESSURE <u>9.031</u> kg/cm²[g] ☐ DIFFERENTIAL PRESSURE <u>7.72</u> kg/cm² ☐ DIFF. HEAD <u>86.1</u> m NPSHA <u>6.38</u> m ☐ PROCESS VARIATIONS _____ ☐ STARTING CONDITIONS _____ SERVICE: ☒ CONT. ☐ INTERMITTENT (STARTS/DAY) _____ ☐ PARALLEL OPERATION REQ'D _____	16
	SITE AND UTILITY DATA	17
	LOCATION: ☐ INDOOR ☐ HEATED ☐ UNDER ROOF ☒ OUTDOOR ☒ UNHEATED ☐ PARTIAL SIDES ☒ GRADE ☐ MEZZANINE _____ ELECTRIC AREA CLASSIFICATION CL <u>1</u> GR _____ IIA _____ DIV <u>ZONE 2</u> ☐ WINTERIZATION REQ'D _____ ☒ TROPICALIZATION REQ'D.	18
	SEE NOTES ON SHEET 6 *	19
	SITE DATA	20
	ALTITUDE _____ BAROMETER _____ RANGE OF AMBIENT TEMPS: MIN/MAX. _____ / _____ RELATIVE HUMIDITY: MIN / MAX _____ / _____ % UNUSUAL CONDITIONS: ☐ DUST ☐ FUMES OTHER _____ UTILITY CONDITIONS:	21
	STEAM: DRIVERS HEATING MIN _____ MAX _____ ELECTRICITY DRIVERS HEATING CONTROL SHUTDOWN VOLTAGE _____ HERTZ _____ PHASE _____ COOLING WATER: TEMP. INLET _____ MAX. RETURN _____ PRESS. NORM _____ DESIGN _____ MIN RETURN _____ MAX ALLOW DP _____ FOULING FACTOR _____	22
	SITE AND UTILITY DATA (CONT'D)	23
	WATER SOURCE _____ CHLORIDE CONCENTRATION (PPM) _____ INSTRUMENT AIR: MAX/MIN PRESS _____ / _____ 0	24
	LIQUID	25
	☐ TYPE OR NAME OF LIQUID <u>Acetone, Phenol, Cumene, CHP</u> ☐ PUMPING TEMPERATURE NORMAL <u>79</u> °C MAX. _____ °C MIN. _____ ☐ VAPOR PRESSURE <u>1.74</u> kg/cm²[a] @ <u>79</u> °C ☐ RELATIVE DENSITY (SPECIFIC GRAVITY): NORMAL <u>0.898</u> MAX _____ MIN _____ ☐ SPECIFIC HEAT _____ 0.51 kcal/(kg.°C) ☐ VISCOSITY <u>0.571</u> cP @ <u>79</u> °C ☐ MAX. VISCOSITY _____ (cP) ☐ CORROSIVE/EROSIVE AGENT _____ ☐ CHLORIDE CONCENTRATION (PPM) _____ ☐ H₂S CONCENTRATION (PPM) _____ LIQUID ☐ HAZARDOUS ☐ FLAMMABLE ☐ OTHER <u>42 mol% Acetone, 42 mol% Phenol, 8.5 mol% Cumene</u> <u>4 mol% Water, 1.5 mol% CHP</u>	26
	PERFORMANCE	27
	PROPOSAL CURVE NO. _____ RPM <u>2950</u> ☐ IMPELLER DIA. RATED _____ MAX. _____ MIN. _____ mm ☐ RATED POWER <u>23.8</u> kW EFFICIENCY <u>66.5</u> % ☐ MINIMUM CONTINUOUS FLOW: THERMAL _____ m³/h STABLE _____ m³/h ☐ PREFERRED OPERATING REGION _____ TO _____ m³/h ☐ ALLOWABLE OPERATING REGION _____ TO _____ m³/h ☐ MAX HEAD @ RATED IMPELLER _____ m ☐ MAX POWER @ RATED IMPELLER _____ kW ☐ NPSHR AT RATED CAPACITY _____ m ☒ SUCTION SPECIFIC SPEED _____ ☐ MAX. SOUND PRESS. LEVEL REQ'D _____ (dBA) ☐ EST MAX SOUND PRESS. LEVEL _____ (dBA)	28
	REMARKS: See Pump Notes 2, 6, 9, C1, C2 and 42.	29
	ESTIMATED SHUTOFF PRESSURE <u>21.00</u> kg/cm²[g]	30

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CENTRIFUGAL PUMPS AND DRIVERS

PROJECT SPECIFICATION

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REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	09-Apr-12	MW	DRL				

REV	CONSTRUCTION	CONSTRUCTION (CONT)	1																					
	APPLICABLE STANDARD: ☒ API 610 11TH EDITION ☐ OTHER _____ (SEE REMARKS)	☐ SHAFT DIAMETER BETWEEN BEARINGS _____ mm ☐ SPAN BETWEEN BEARING CENTERS _____ mm ☐ SPAN BETWEEN BEARING & IMPELLER _____ mm	2																					
	PUMP TYPE: ☒ OH2 ☐ BB1 ☐ VS1 ☐ VS6 ☐ OH3 ☐ BB2 ☐ VS2 ☐ VS7 ☐ OH4 ☐ BB3 ☐ VS3 ☐ OTHER ☐ OH5 ☐ BB4 ☐ VS4 ☐ OH6 ☐ BB5 ☐ VS5	REMARKS: COUPLINGS: _____ DRIVER - PUMP ☐ MAKE _____ ☒ MODEL _____ ☐ RATING _____ ☐ LUBRICATION _____ ☒ LIMITED END FLOAT REQUIRED _____ ☒ SPACER LENGTH _____ mm ☒ SERVICE FACTOR _____	3																					
	NOZZLE CONDITIONS: <table border="1" style="width:100%; border-collapse: collapse;"> <thead> <tr> <th>SIZE (IN)</th> <th>FLANGE CLASS</th> <th>FAC'G</th> <th>POSITION</th> </tr> </thead> <tbody> <tr> <td>SUCTION</td> <td>300</td> <td>RF</td> <td></td> </tr> <tr> <td>DISCHARGE</td> <td>300</td> <td>RF</td> <td></td> </tr> <tr> <td>BALANCE DRUM</td> <td></td> <td></td> <td></td> </tr> </tbody> </table>	SIZE (IN)	FLANGE CLASS	FAC'G	POSITION	SUCTION	300	RF		DISCHARGE	300	RF		BALANCE DRUM				DRIVER HALF COUPLING MOUNTED BY: ☐ PUMP MFR. ☐ DRIVER MFR. ☐ PURCHASER ☐ COUPLING PER API 671	4					
SIZE (IN)	FLANGE CLASS	FAC'G	POSITION																					
SUCTION	300	RF																						
DISCHARGE	300	RF																						
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	PRESSURE CASING CONNECTIONS: <table border="1" style="width:100%; border-collapse: collapse;"> <thead> <tr> <th>NO.</th> <th>SIZE (NPS)</th> <th>TYPE</th> </tr> </thead> <tbody> <tr> <td>☐ DRAIN</td> <td></td> <td>Provide With Valve</td> </tr> <tr> <td>☐ VENT</td> <td></td> <td></td> </tr> <tr> <td>☐ PRESSURE GAUGE</td> <td></td> <td></td> </tr> <tr> <td>☐ TEMP GAUGE</td> <td></td> <td></td> </tr> <tr> <td>☐ WARM-UP</td> <td></td> <td></td> </tr> <tr> <td>☐ BALANCE / LEAK-OFF</td> <td></td> <td></td> </tr> </tbody> </table>	NO.	SIZE (NPS)	TYPE	☐ DRAIN		Provide With Valve	☐ VENT			☐ PRESSURE GAUGE			☐ TEMP GAUGE			☐ WARM-UP			☐ BALANCE / LEAK-OFF			BASEPLATES: API BASEPLATE NUMBER _____ (APPENDIX M) ☐ NON-GROUT CONSTRUCTION:	5
NO.	SIZE (NPS)	TYPE																						
☐ DRAIN		Provide With Valve																						
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☐ PRESSURE GAUGE																								
☐ TEMP GAUGE																								
☐ WARM-UP																								
☐ BALANCE / LEAK-OFF																								
	☐ CYLINDRICAL THREADS REQUIRED	REMARKS: MATERIAL ☐ APPENDIX H CLASS _____ A-8 ☐ MIN DESIGN METAL TEMP _____ 15 °C ☐ BARREL/CASE 316 SS IMPELLER 316 SS ☐ CASE/IMPELLER WEAR RINGS Hard-Faced 316 SS ☐ SHAFT 316 SS ☐ DIFFUSERS _____ ☐ COUPLING SPACER/HUBS _____ ☐ COUPLING DIAPHRAGMS (DISKS) _____	6																					
	CASING MOUNTING: (SEE SEPARATE SHEET FOR VERTICALS) ☒ CENTERLINE ☐ NEAR CENTERLINE ☐ FOOT ☐ SEPARATE MOUNTING PLATE ☐ IN-LINE		7																					
	CASING SPLIT: ☐ AXIAL ☒ RADIAL		8																					
	CASING TYPE: ☐ SINGLE VOLUTE ☒ MULTIPLE VOLUTE ☐ DIFFUSER ☒ OVERHUNG ☐ BETWEEN BEARINGS ☐ BARREL		9																					
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	ROTATION: (VIEWED FROM COUPLING END) ☐ CW ☐ CCW ☐ IMPELLERS INDIVIDUALLY SECURED		11																					
	REMARKS: ☐ BOLT OH3 PUMP TO PAD/FOUNDATION SHAFT: ☐ SHAFT DIAMETER AT COUPLING _____ mm		12																					
	REMARKS: SUCTION TYPE <u>Single</u> IMPELLER TYPE <u>Closed</u> Pumps shall comply with UOP Standard Specifications 5-11 and 7-12 See Project Specification 533.		13																					
		BEARINGS AND LUBRICATION BEARING (TYPE/NUMBER): ☐ RADIAL _____ Ball _____ / _____ ☐ THRUST _____ Ball _____ / _____ ☐ REVIEW AND APPROVE THRUST BEARING SIZE LUBRICATION: ☒ GREASE ☒ FLOOD ☐ RING OIL ☒ FLINGER ☐ PURGE OIL MIST ☒ PURE OIL MIST ☐ CONSTANT LEVEL OILER PREFERENCE (SEE REMARKS) ☐ PRESSURE LUBE SYS ☐ API-610 ☐ API-614 ☒ OIL VISC. ISO GRADE _____	14																					

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0	09-Apr-12	MW	DRL				
1	13-Aug-12	MW	DRL				

REV

NOTES:

* SEE BASIC ENGINEERING DESIGN QUESTIONNAIRE (BEDQ), ENGINEERING DESIGN INFORMATION (EDI) PROJECT 963763 A.4, SECTIONS 3 AND 4.

The following notes apply to those services for which they are specified in this Project Specification:

PUMP NOTES:

2. Pump vendor shall provide non-contacting pressurized dry gas seals. The barrier gas shall be Nitrogen from the plant Nitrogen header available at 7 kg/cm²(g). Pump vendor shall provide a seal control panel for each pump. The panel shall include the following:

- Coalescing filter (removal efficiency of 98% on liquids 1 micron or greater and solids 3 microns or greater)
- Pressure regulating valve
- Pressure gauge
- Flow meter low range
- Flow meter high range

Alarm switches:

- Low Pressure
- High Flow

The alarm switches shall activate a control center trouble alarm (by others). The contractor and pump vendor shall coordinate switch selection.

A mechanical amplifier shall be provided by the seal vendor to boost the nitrogen buffer supply pressure when the normal nitrogen supply pressure does not exceed either:

- the pump suction pressure by 2.1 kg/cm², or
- the pump maximum suction pressure by 0.35 kg/cm² psi.

When these conditions exist, the seal gas control panel scope shall be expanded to include the amplifier and a minimum 2 gallon stainless steel accumulator vessel with local pressure gauge.

6. Mechanical seal parts and packings shall be suitable for handling aromatic hydrocarbons.

9. The "capacity, other" is based on 50% of the lowest normal operating flow.

11. Pumps shall be mounted on cover plates to be furnished by vessel supplier to suit sump tank opening. Each pump shall include a stainless steel suction screen. A steel discharge pipe shall extend through the cover plate terminating in a Cl. 300 flanged connection. Contractor to determine depth of setting. Pump will be spared by common warehouse spare pump P-1304S.

42. VOC packing valves should be supplied.

C1. Casing volume to be minimized and subjected to critical volume restriction

C2. Cumene Hydroperoxide is thermally unstable that can initiate self-heating reaction at elevated temperatures

C3. Acid Aromatics sump pump P-2304A shall be identical to the existing Acid Aromatics sump pump in PPCL Unit 1 (Item No. P-1304A/S). Existing warehouse spare pump P-1304S shall be used as a common spare for both services.

MOTOR NOTES:

M5. Non Overloading at all points on pump curve

M10. Motor shall have torque capability to accelerate the pump to operating speed with discharge shutoff valve open, and the discharge check valve closed against a static pressure of 1.8 kg/cm² (g)

M12. Deleted.

M13. The standby pump motor shall be connected to an alternate independent power supply.

GENERAL NOTES:

G1. Lubrication system is applied to high voltage motor's and all pumps' bearings except for vertical and diaphragm pumps.

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